

EDUCATION

North Carolina State University (NCSU)

Ph.D. Computer Science (CGPA: 4.0/4.0), Advisor: Dr. Anupam Das

Ph.D. Proposal Defense: Passed

MS Computer Science (CGPA: 4.0/4.0)

Raleigh, North Carolina

Aug 2021 — Oct 2026 (Expected)

December 2025

December 2024

Lahore University of Management Sciences (LUMS)

BS Computer Science (CGPA 3.57/4.0)

Lahore, Pakistan

Aug 2016 — May 2020

EXPERIENCE

Wolfpack Security and Privacy Research (WSPR) Lab @ North Carolina State University

Raleigh, North Carolina

Graduate Research Assistant (PhD Candidate)

Aug 2021 — Present

- I conduct quantitative and qualitative research in the area of privacy and security of Smart Voice Assistants. I design privacy interventions to bridge the gap between users' mental models and actual system behavior, enabling users to make informed decisions about sharing personal information.
- Impact:** 10+ peer-reviewed publications in top-tier privacy and security conferences with 150+ citations. Bug bounty from Amazon for vulnerability disclosure.
- Research Interests:** Investigating Socio-technical Problems, Algorithmic Auditing, User Mental Model Studies, Designing Privacy Interventions

Allthenticate

Houston, Texas

Software Engineer (Summer Intern)

May — Aug 2023

- As a summer SWE intern (Full-stack Software Engineer), I developed components of the admin portal built on Flask, Vue 3, MongoDB (for system logs), and SQL. Updated SQL schema to support new features and optimized query performance.
- Improved the event logging by increasing granular controls coverage by 50%.
- Analyzed logs and estimated the efficiency of the company's products (for research and analytics).

Educative.io

Lahore, Pakistan

Technical Content Engineer

Aug 2020 — Sep 2021

- I created Vue.js course [<https://www.educative.io/courses/hands-on-vue-js-build-fully-functional-spa>] along with 5 other courses on programming languages.
- Collaborated on a project with Wix to create a Wix website builder hands-on experience on the Educative platform.

RESEARCH PUBLICATIONS

- Privacy by Voice: Designing Usable Privacy Interventions for the Voice Interface. Accepted at *PETS 2026*. [Aafaq Sabir](#), Abhinaya S.B., Dilawer Ahmed, and Anupam Das.
- PriVA-C: Defending Voice Assistants from Fingerprinting Attacks. *PETS 2026*. **(Received Best Paper Runner-up Award)** Dilawer Ahmed, [Aafaq Sabir](#), Ahsan Zafar, and Anupam Das.
- Analyzing Ad Prevalence, Characteristics, and Compliance in Alexa Skills. *IEEE S&P 2025*. [Aafaq Sabir](#), Abhinaya S.B., Dilawer Ahmed, and Anupam Das.
- Spying through your voice assistants: realistic voice command fingerprinting. *USENIX Security 2023*. Dilawer Ahmed, [Aafaq Sabir](#), and Anupam Das.
- Speaker Orientation-Aware Privacy Control to Thwart Misactivation of Voice Assistants. *DSN 2023*. Shaohu Zhang, [Aafaq Sabir](#), and Anupam Das.
- Hey Alexa, Who am I talking to?: Analyzing Users' Perception and Awareness regarding Third-party Alexa Skills. *CHI 2022*. [Aafaq Sabir](#), Evan Lafontaine, and Anupam Das.
- Analyzing the Impact and Accuracy of Facebook Activity on Facebook's Ad-Interest Inference Process. *CSCW 2022*. [Aafaq Sabir](#), Evan Lafontaine, and Anupam Das.
- Understanding the Privacy Implications of Adblock Plus's Acceptable Ads. *ASIACCS 2021*. Ahsan Zafar, [Aafaq Sabir](#), Dilawer Ahmed, and Anupam Das.

PAPER PRESENTATIONS

- Presented the paper "Analyzing Ad Prevalence, Characteristics, and Compliance in Alexa Skills" at 2025 IEEE Symposium on Security and Privacy (SP), San Francisco, CA, USA. **Available online:** <https://www.youtube.com/watch?v=Kh6FaZ2Cx2w>

- Presented the paper “Hey Alexa, Who Am I Talking to?: Analyzing Users’ Perception and Awareness Regarding Third-party Alexa Skills” at 2022 CHI Conference on Human Factors in Computing Systems, New Orleans, LA, USA. **Available online:** <https://www.youtube.com/watch?v=EhLvM2cIQXE>
- Presented the paper “Analyzing the Impact and Accuracy of Facebook Activity on Facebook’s Ad-Interest Inference Process” at 2022 Proceedings of ACM Hum. Comput. Interact. 6, CSCW. **Available online:** <https://www.youtube.com/watch?v=iMjSaFPWNNQ>

POSTER PRESENTATIONS

- Aafaq Sabir, Anupam Das. Automated Interaction With Smart Virtual Assistants Voice Apps Using Large Language Models. Presented at *USENIX Security* ’23.
- Shaohu Zhang, Aafaq Sabir, Anupam Das. Enhancing Security and Privacy Control for Voice Assistants Using Speaker Orientation. Presented at *WiSec* ’23.
- Evan Lafontaine, Aafaq Sabir, Anupam Das. Understanding people’s attitude and concerns towards adopting IoT devices. Presented at *CHI Extended Abstracts* ’21.

TEACHING EXPERIENCE

North Carolina State University (NCSU)	Raleigh, North Carolina
<i>Instructor and Administrator (Summer Semester)</i>	<i>May — July 2025</i>
• Appointed as Instructor and Administrator for CSC 116 Intro to Programming in Java	
<i>Teaching Assistant</i>	<i>Jan — May 2023</i>
• Appointed as TA for CSC 574 (Computer and Network Security): Delivered lecture on Anonymous Communication	

FELLOWSHIPS

North Carolina State University	Raleigh, North Carolina
<i>Mentored Teaching Fellow</i>	<i>Jan — May 2026</i>
• Selected as Mentored Teaching Fellow for the course CSC 433: Privacy in the Digital Age • Responsibilities included teaching (delivering lectures), improving course structure and components, i.e., homework, programming assignments, and projects.	

LEADERSHIP AND OUTREACH ACTIVITIES

In Collaboration with North Carolina School of Science and Mathematics (NCSSM)	Raleigh, North Carolina
<i>Summer Research Mentor. Mentee: Adrita Fuad</i>	<i>May — July 2024</i>
North Carolina State University (NCSU)	Raleigh, North Carolina
<i>BS Independent Research Mentor. Mentee: Divya Srinivasan</i>	<i>September 2023 — April 2024</i>
North Carolina State University (NCSU)	Raleigh, North Carolina
<i>MS Independent Research Mentor. Mentee: Pratik Devnani</i>	<i>January 2023 — May 2023</i>
In Collaboration with North Carolina School of Science and Mathematics (NCSSM)	Raleigh, North Carolina
<i>Summer Research Mentor. Mentees: Jonathan Liang, Eric Ge</i>	<i>May — July 2022</i>
In Collaboration with King AbdulAziz University (KAU)	Raleigh, North Carolina
<i>Summer Research Mentor. Mentees: Abdulaziz Altamimi, Jasser Al-Ahmadi</i>	<i>May — July 2022</i>

AWARDS

- Recipient of College of Engineering CSC Department Travel Award 2023 of \$1000 for Conference Travel Reimbursement (IEEE S&P 2025).
- Recipient of College of Engineering CSC Department Travel Award 2023 of \$1000 for Conference Travel Reimbursement (USENIX Security 2023).
- Recipient of SIGCHI Gary Marsden Travel Awards 2022 of \$700 for Conference Travel Reimbursement (CHI 2022).
- Recipient of College of Engineering CSC Department Travel Award 2022 of \$1000 for Conference Travel Reimbursement (CHI 2022).

SERVICE

- Invited as a Program Committee member at USENIX Security 2027.
- Program Committee member: The 20th International AAAI Conference on Web and Social Media, ICWSM 2026.
- Invited reviewer: The 29th ACM Conference on Computer-Supported Cooperative Work and Social Computing, CSCW 2026.

SKILLS

- **Analytics/Research:** Fine-tuning LLMs (proprietary and open-source using LoRA/PEFT), RAG, Context/Prompt Optimization, Agent Orchestration, LangChain, ML (TensorFlow), Prompt Engineering and Evaluation, OpenAI and Anthropic APIs, LLM Guardrails, ETL Pipelines, Statistical Modeling (regression, multivariate methods), Qualitative and Quantitative Methods, Applied Data Science
- **Cloud computing:** AWS EC2, GCP, Local org cloud servers (Proxmox)
- **Dev stack:** Node.js, Vue.js, Express, MongoDB, Browser Extension Dev., Android Dev.
- **Programming languages/Tools:** C++, C, Python, R, JavaScript, Golang, Java, SQL, REST API, AWS services (EC2 and S3), Docker, Multi-threaded apps, client-server programming

BUG BOUNTIES

- Received Bug Bounty in **Amazon Vulnerability Research Program - Devices (2024)**

SOFTWARE ENGINEERING PROJECTS

CheckMate: Built an online platform to conduct a coding competition. The web app runs the submitted code (currently supports C++), evaluates the results based on the provided test cases, and updates the leaderboard.

AnonQ: Built an anonymous question answering platform with MERN (MongoDB, Express.js, React, Node.js) stack with the functionality of comment replies. GitHub: <https://github.com/Aafaq52/anonq>.

Deployed at Vercel: <https://anonq-umber.vercel.app/>

GRANTS

North Carolina State University (NCSSU)

Raleigh, North Carolina

Assisted Dr. Anupam Das (PhD Advisor) with research proposals and grant writing.

NOTABLE COURSES

Grad: Computer and Network Security, Cryptography, OS Security, Adv. Internet Protocols, Compiler Construction

Undergrad: Artificial Intelligence, Network Security, Software Engineering, NLP, Operating systems, Algorithms

REFERENCES

Dr. Anupam Das (anupam.das@ncsu.edu), Dr. Brad Reaves (bgreaves@ncsu.edu),

Dr. Dominik Wermke (dwermke@ncsu.edu)